

Security-Aware Workload-State Counterfactual Verification and Nodal Net-Value Settlement for Spatially Coupled AI Data Centers

Anonymous Authors

Abstract—AI data centers can provide demand response (DR), yet a meter decrease does not show that the workload was committed before the event or that the change had positive network value. This paper develops a declaration-indexed verifier and settlement chain for spatially coupled data centers. Submit-time scheduler records define releases, runtimes, requested graphics processing units (GPUs), classes, and regions. A cumulative workload linear program (LP), a validation-fitted convex simplex, and an exact contiguous start-time witness share that ledger and its digest; execution telemetry is opened only for an independent replay. The simplex enforces total and daily conditional value at risk (CVaR) limits on absolute false-credit exposure and reports independent Karush–Kuhn–Tucker and primal residuals. On 54 locked days, the joint fit has normalized root-mean-square error (nRMSE) 1.123, false credit 0.303 MWh/day, and F1 score (harmonic mean of precision and recall) 0.373; the single feasible projection has 0.943, 2.470 MWh/day, and 0.463, respectively. On the independent meter replay, the joint fit has nRMSE 0.455 versus 0.716 for that projection. Its validation certificate reduces false exposure from 58.104 to 4.507 MW-slots under a 14,526 MW-slot total budget and reduces the 75th-percentile daily tail from 12.993 to 0.966 under a 3.248 budget. A separate gate experiment removes post-gate arrivals, requires 3 MWh of committed event service, and verifies all 54 locked days. The resulting evidence distinguishes declaration feasibility, simulated mechanism isolation, and observational scoring; it does not identify a causal utility treatment effect.

Index Terms—AI data centers, counterfactual verification, demand response, security-constrained dispatch, nodal settlement.

I. INTRODUCTION

The growth of large-model training and inference is converting data centers from nearly static industrial loads into large, software-defined grid resources. Interactive inference is latency constrained, elastic inference can be delayed briefly, and batch GPU work can be deferred for hours or migrated between regions. These capabilities enable demand response (DR), but they also invalidate the usual interpretation of a meter decrease. A reduction at one facility can be followed by delayed consumption or simultaneous rebound at another node. Paying the positive local decrease can therefore reward a redistribution that creates little or negative network value. Public traces already expose multi-week bursty workload arrivals and tens of thousands of scheduler jobs, while facility telemetry reports GPU energy at a different measurement boundary [1], [2].

Conventional DR measurement and verification estimates the unobserved no-event demand from historical meter data.

Arithmetic historical baselines remain common in market practice under published market guidance [3], despite known bias and manipulation incentives [4]. Regression-based facility models quantify baseline uncertainty but do not encode a computing workload ledger [4]. More broadly, DR research establishes the economic role of responsive load, yet a meter-only estimator cannot determine whether queued jobs could actually have produced its predicted trajectory.

Data-center scheduling studies provide this physical state: temporal deferral and geographical balancing reduce energy cost and peak demand [5]. Data-driven flexibility and security-constrained coordination optimize spatial-temporal dispatch [5], [6]. Balancing-market and load-shifting studies provide complementary dispatch contexts. Virtual-link clearing and space-time remuneration model geographical and temporal transfers [7], [8]. Receding-horizon frequency regulation retains multi-site state and future arrivals [9]. These methods optimize operator-controlled schedules; verification of an independently submitted DR counterfactual against an auditable workload ledger remains open. Coupled-regulation models likewise provide no provenance-bound certificate [10]. Whole-facility measurements also require a bottom-up GPU-to-facility conversion; calibration, benchmark scaling, and network placement are therefore kept as separate evidence layers.

The second gap is electrical. Geographic workload control is often valued with energy prices or gross megawatts, although nodal marginal value changes when congestion or a contingency binds. Direct-current (DC) power flow and linear distribution factors provide transparent sensitivities [11], [12]; N-1 dispatch imposes credible post-contingency limits through line-outage distribution factors (LODFs) [13]. No data-center DR verifier jointly certifies workload feasibility, signed spatial response, and security-aware network value. Alternating-current (AC) feasibility is a separate nonlinear check. Public traces support observational replay but not a causal utility event, so measured execution, counterfactual construction, and network valuation must remain separate.

This paper makes three power-system contributions. 1) It defines a typed submit-execute-settle evidence chain in which a runtime-complete job witness is formed only from scheduler declarations (release, allocation runtime, requested GPUs, and regional assignment). The same indexed witness digest is rechecked at regional aggregation and reused verbatim by the RTS-24 N-1 value and full-day settlement replay;

execution telemetry is retained for post-event scoring only. 2) It formulates a workload-state verifier with an explicit information boundary and a validation-fitted total-plus-daily-CVaR risk fit, then gives that aggregate target an exact fixed-price contiguous-block realization under the same declaration digest. The central calibrated entitlement and the nameplate-capacity upper commitment are both certified instead of being conflated. 3) It develops signed space-time nodal remuneration and a payment-activation rule over calibration-validation power-conversion factors: the raw q_{99} endpoint is retained under the declared fixed/flexible payment-network scale, while the independently reconciled flexible-only capacity factor is reported as a planning diagnostic and never used to clip that endpoint. The central result is a dimension-preserving coupling invariant that carries one declared workload state into physical verification, network valuation, and settlement; the linear-program (LP), dispatch, and allocation primitives retain their established power-system semantics. Because public traces do not identify utility geography, spatial conclusions are conditional on the declared feature-stratified trace scenario and the complete permutation/penetration panel. The certificate assumes trusted scheduler and telemetry provenance: a digest detects post-commitment record changes but cannot authenticate semantically false fields.

The remainder of the paper is organized as follows. Section II defines the spatial workload and network settlement problem. Section III presents the verifier, security-aware value rule, and allocation mechanism. Section IV specifies the data, baselines, and locked protocol. Section V reports the core counterfactual, execution-replay, job-level, security, payment, continuous-horizon, and provenance experiments. Section VI concludes with the deployment boundary and future validation requirements.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. Spatial workload state

Let $\mathcal{S}$, $\mathcal{K}$, $\mathcal{D}$, and $\mathcal{T}$ denote source regions, workload classes, destination data centers, and 15-minute intervals; the symbol h is reserved for a calendar day in statistical panels. The submit-time declaration ledger supplies arrival energy a_{skt} for source s , class k , and release interval t . The decision $x_{skd\tau} \geq 0$ is the energy served at destination d in interval τ . Its cumulative state is

$$y_{skt} = \sum_{\tau=0}^t \sum_{d \in \mathcal{D}} x_{skd\tau}. \quad (1)$$

With cumulative arrivals $A_{skt} = \sum_{\tau=0}^t a_{sk\tau}$ and deadline D_k , release and deadline feasibility are

$$y_{skt} \leq A_{skt}, \quad (2)$$

$$y_{skt} \geq A_{sk,t-D_k}, \quad t \geq D_k. \quad (3)$$

The state recursion is $y_{skt} - y_{sk,t-1} = \sum_d x_{skd\tau}$ with $y_{sk,-1} = 0$, and terminal service equals the declared arrivals due in the window; the release and deadline inequalities are (2) and (3), and the site-capacity constraint is $\sum_{s,k} x_{skd\tau} \leq \bar{P}_d \Delta t$. Facility demand is

$$p_{dt} = P_d^{\text{fix}} + \Delta t^{-1} \sum_{s,k} x_{skd\tau}. \quad (4)$$

For rolling window end T^+ and final event-day release T^0 , the terminal state is not forced to consume unexpired future work. Instead,

$$y_{skT^+} = A_{sk,\max\{T^0, T^+ - D_k\}}, \quad (5)$$

which completes every job due inside the window and every event-day arrival while retaining later real arrivals. The window starts $D_{\max} + 96$ intervals before and ends $D_{\max}$ intervals after the event; omitted earlier work has expired. Equations (2)–(4) map an aggregate preemptive envelope to the facility meter. This is valid for checkpointable batch work; Experiment 14 separately audits contiguous execution intervals, runtime, GPU count, and native power. For the counterfactual task-level check, let $j \in \mathcal{J}_{\text{sub}}$ index a submitted scheduler record, r_j denote its submitted release, and let d_j be the service-window endpoint. Slurm “timelimit” is an allocation run-time declaration; it is converted to $R_j = \lceil \text{timelimit}_j / \Delta t \rceil$ slots and combined with a precommitted queue allowance $B = 96$ slots, so $d_j = r_j + R_j + B$. The unlimited Slurm sentinel is mapped to the precommitted horizon $H_{\infty} = 128$ slots before adding B ; finite declarations are retained exactly. The 0.001-MW-per-GPU nameplate is fixed before the split. The training-calibrated entitlement $\hat{E}_j$ is retained only for the aggregate Exp19 audit; the executable witness uses the declared runtime. Let $\bar{p}_{\text{GPU}} = 0.001$ MW/GPU be the committed nameplate, s_j the submit-time scenario region, and g_j^{req} the requested GPU count. Define the runtime-complete entitlement $E_j^{\text{run}} = R_j g_j^{\text{req}} \bar{p}_{\text{GPU}} \Delta t$. The indexed service variable $u_{j\tau}$ is the energy assigned to job j in interval τ , and the executable witness requires one contiguous fixed-rate block of exactly R_j slots. Let w_{batch} be the predeclared waiting-cost coefficient per MWh-slot, and r^{DR} the event service tariff. Define $c_j = g_j^{\text{req}} \bar{p}_{\text{GPU}} \Delta t$ and $\Theta_j = \{r_j, \dots, d_j - R_j\}$. For each admissible start θ , a binary $z_{j\theta}$ selects the block and $u_{j\tau}^{\theta} = c_j \mathbf{1}\{\theta \leq \tau < \theta + R_j\}$. The exact start-time mixed-integer linear program (MILP) is

$$\begin{aligned} \sum_{\theta \in \Theta_j} z_{j\theta} &= 1, \quad z_{j\theta} \in \{0, 1\}, \\ u_{j\tau} &= \sum_{\theta \in \Theta_j} u_{\tau}^{\theta} z_{j\theta}, \quad \sum_{j:s_j=d} u_{j\tau} \leq \bar{P}_d \Delta t, \\ \min_z \sum_{j,\theta \in \Theta_j} z_{j\theta} \sum_{\tau} [w_{\text{batch}}(\tau - r_j) + r^{\text{DR}} \mathbf{1}\{\tau \in \mathcal{E}\}] u_{\tau}^{\theta}. \end{aligned} \quad (6)$$

Every selected block enforces runtime completion, contiguity, and the GPU nameplate; its positive regional slack makes the separable enumeration exact for the displayed MILP. For delay $\delta = \theta - r_j$, the waiting part of the objective is $w_{\text{batch}} c_j \{R_j \delta + R_j(R_j - 1)/2\}$, so a long job is charged for every delayed service interval rather than only for its start time. Experiment 27 enumerates every admissible start for all 75,326 records and stores the earliest-start baseline and tariff response; Experiment 25 adds a binding regional row. The measured $[t_j^{\text{start}}, t_j^{\text{end}}]$ interval is opened only for independent post-event scoring after the declaration witness is fixed: 71,128 execution matches never constrain z , u , or d_j . Thus $\hat{E}_j$ is calibration-only, E_j^{run} is the nameplate-capacity commitment, and the calibrated central energy is carried on the same blocks without being presented as measured consumption. The risk

target is linked through a fixed regional slot price. Because the statistical and nameplate ledgers have different scales, define the executable contract $\tilde{p}_{dht}^{\text{risk}} = P_d^{\text{fix}} + \gamma_+ [\tilde{p}_{dht}^{\text{risk}} - P_d^{\text{fix}}]_+$. Here $\gamma_+ = E_{\text{lock}}^{\text{submit},+} / \sum_{d,h,t} [\tilde{p}_{dht}^{\text{risk}} - P_d^{\text{fix}}]_+ \Delta t$ is computed once from submit-time nameplate declarations on the locked days. For locked day h , region d , and interval t , define $\pi_{dht} = r^{\text{DR}} [\tilde{p}_{dht}^{\text{risk}} - P_d^{\text{fix}}]_+ / \tilde{\Pi}_{\max}$, where $\tilde{\Pi}_{\max} = \max_{d,h,t} [\tilde{p}_{dht}^{\text{risk}} - P_d^{\text{fix}}]_+$. Re-enumerating every declaration-feasible block with the waiting, event-tariff, and π_{dht} terms produces the exact risk-priced realization. The bridge stores raw/scaled profiles, the common digest, daily residuals, and the central-energy check; it is a deterministic realization of the fitted target, not a second aggregate optimization. For the job-to-network certificate, the indexed witness is mapped without a second optimization. To avoid overloading the deadline endpoint d_j , let $\mathcal{J}_{sk\delta}$ be the ledger partition whose source, class, and declared destination labels are s, k, δ ; Experiment 27 uses the native assignment $\delta = s_j$ and therefore has no outcome-dependent migration:

$$x_{sk\delta\tau}^{\text{job}} = \sum_{j \in \mathcal{J}_{sk\delta}: r_j \leq \tau < d_j} u_{j\tau}, \quad (7)$$

$$p_{\delta\tau}^{\text{job}} = P_\delta^{\text{fix}} + \Delta t^{-1} X_{\delta\tau}^{\text{job}}, \quad X_{\delta\tau}^{\text{job}} = \sum_{s,k} x_{sk\delta\tau}^{\text{job}}. \quad (8)$$

Aggregate migration remains a matched-workload control; (8) is the indexed replay profile. Experiment 27 reconstructs X^{job} from every stored $u_{j\tau}$ and verifies equality before network valuation; no separately optimized aggregate receives a network value. Workload deferral and geographic assignment follow [5], [6], while rolling-state control follows [9]; waiting and migration costs are $w_k(\tau - t)x_{skd\tau}$ and $m_{sd}x_{skd\tau}$, with service-time substitution differing only by a fixed-arrival constant.

The submit-time and post-event ledgers have separate canonical digests. Each submitted record is represented by the scheduler-only tuple; its Secure Hash Algorithm 256 (SHA-256) digest is

$$\begin{aligned} \ell_j^{\text{sub}} &= (\text{id}_j, t_j^{\text{submit}}, \text{timelimit}_j, g_j^{\text{req}}, \kappa_j, \sigma_j), \\ \mathcal{L}_{\text{sub}} &= \text{sort}\{\ell_j^{\text{sub}} : j \in \mathcal{J}_{\text{sub}}\}, \\ h_{\text{sub}} &= \text{SHA-256}(\text{serialize}(\mathcal{L}_{\text{sub}})). \end{aligned} \quad (9)$$

where g_j^{req} , κ_j , and σ_j are the requested-GPU, class, and scheduler-state fields. The resulting canonical commitment is (9). Start/end times, DCGM energy, and telemetry counts enter a separate execution ledger whose join and energy equality are checked only after commitment. Canonical ordering makes h_{sub} independent of execution telemetry [14]; Experiment 16 reconciles measured power with the 118-MW nameplate fixed before the split.

The verifier assumes scheduler/DCGM services are authoritative and the digest is committed before the event decision. It cannot establish that a deadline, region, class, or energy field was truthful before signing; upstream attestation is outside this certificate.

B. Counterfactual, response, and false credit

For event set $\mathcal{E} \subset \mathcal{T}$, p denotes regional power in MW and time sums use Δt (MWh). The declaration-causal estimate is $\tilde{p}$; p^0 is the declaration-only strategic reference used for

mechanism isolation, $p^{1,\text{sim}}$ is the independently solved event-response schedule, and p^{obs} is the Massachusetts Institute of Technology (MIT) Data Center GPU Manager (DCGM) execution trace. The observed trace is opened only after a profile is frozen; no utility event label exists, so it is never a causal input. The mechanism-isolation response $r = p^0 - p^{1,\text{sim}}$ is signed and retains rebound. For $z \in \{\text{obs}, \text{sim}\}$, submitted and true credit are $\hat{r}^{(z)} = [\tilde{p} - p^{(z)}]_+$ and $r^{(z)} = [p^0 - p^{(z)}]_+$; false credit is

$$F^{(z)} = \Delta t \sum_{d,t \in \mathcal{E}} [\hat{r}_{dt}^{(z)} - r_{dt}^{(z)}]_+, \quad z \in \{\text{obs}, \text{sim}\}. \quad (10)$$

The simulated source defines the validation epigraph; the observed source is recomputed only in locked replay. False credit is computed from (10) in absolute MW-slots, avoiding a ratio with a zero-credit denominator. Before the event the operator freezes a contract baseline p^{con} . After meter closure, define the submitted gross reduction, the contract-cap reduction, and the payable reduction by

$$\begin{aligned} q_{dt}^{\text{gross}} &= [\tilde{p}_{dt} - p_{dt}^{\text{obs}}]_+, \\ q_{dt}^{\text{cap}} &= [p_{dt}^{\text{con}} - p_{dt}^{\text{obs}}]_+, \\ q_{dt}^{\text{pay}} &= \min\{q_{dt}^{\text{gross}}, q_{dt}^{\text{cap}}\}. \end{aligned} \quad (11)$$

The capacity payment is $Q^{\text{pay}} = \Delta t \sum_{d,t \in \mathcal{E}} q_{dt}^{\text{pay}}$. Equation (11) uses only the submitted baseline, frozen contract, and closed meter. Here $p^{\text{con}} = p^{\text{safe}}$ for the payment-band panel and is the no-tariff committed-ledger profile for the decision-time panel; p^0 remains an offline diagnostic. On locked event slots, $\text{nRMSE} = \sqrt{N_E^{-1} \sum_n (\tilde{p}_n - p_n^0)^2 / \max\{N_E^{-1} \sum_n p_n^0, \epsilon\}}$. With $M = \sum_n \min\{\hat{r}_n, r_n\}$, precision, recall, and F_1 are $M / \max\{\sum_n \hat{r}_n, \epsilon\}$, $M / \max\{\sum_n r_n, \epsilon\}$, and $2PR / \max\{P + R, \epsilon\}$ over the N_E region-slot samples. The reported error metrics use mean absolute error (MAE) and root-mean-square error (RMSE); normalized RMSE (nRMSE) is the scale-free form above. The risk simplex is the sole convex quadratic program (QP); all released workload counterfactuals and payment caps use the linear programs stated above.

C. Strategic reference scheduling

Consider an arithmetic mean of $H = 10$ reference days, event probability q , and DR price π^{DR} . If reference day j adds event-window service δR_j , the expected payment change is

$$\Delta \Pi = \frac{q\pi^{\text{DR}}}{H} \sum_{j=1}^H \delta R_j - \sum_{j=1}^H \Delta C_j. \quad (12)$$

Let c'_j be the right derivative of minimum operating cost with respect to the event-window service constraint on day j . Linear-program sensitivity [15] gives

$$\frac{q\pi^{\text{DR}}}{H} > \min_j c'_j \iff \exists \text{ a strictly profitable local deviation.} \quad (13)$$

At a degenerate optimum the current dual can be zero while the right marginal cost is positive; Experiment 1 therefore checks c'_j with independent 5×10^{-4} and 10^{-3} MWh finite differences (agreement tolerance 10^{-6} USD/MWh). Equations (12)–(13) specialize the historical-baseline premise in [4].

D. Network value

Let M be the data-center-to-bus incidence matrix and $p_t = p_t^{\text{native}} + Mp_t^{\text{dc}}$ the nodal demand vector. For generator-to-bus map C_g , generator bounds, and power-transfer distribution factor (PTDF) H with line ratings $\bar{f}$, the piecewise-linear security-constrained economic dispatch (SCED) value is

$$V(p) = \min_g \sum_i C_i(g_i) \quad (14)$$

$$\text{s.t. } \mathbf{1}^\top g = \mathbf{1}^\top p, \quad \underline{g} \leq g \leq \bar{g},$$

$$-\bar{f} \leq H(C_g g - p) \leq \bar{f}.$$

The incidence mapping aggregates demand at its connection bus; (14) is the standard lossless DC representation [11], [12]. The realized signed grid value is

$$\Delta V = V(p^0) - V(p^1). \quad (15)$$

Positive ΔV is a credit and negative ΔV a debit; no positive-only truncation is applied.

III. PROPOSED METHODOLOGY

A. Model-based framework and matched-information pre-estimation

The state, power, and network equations define the feasible state before a payment target is formed (Fig. 1). Meter history and submitted declarations enter the estimator in parallel; execution remains scoring-only. Lagged meter/calendar baselines and declaration-view ledger comparators include ridge, gradient boosting, and extremely randomized trees [16], [17]. Synthetic control and nonnegative donor weighting are evaluated as a separate history-based class [18], [19]. The quantile output receives the same exact projection, with its penalty selected by contiguous folds.

B. Sparse cumulative workload-state verifier

For statistical estimate $\tilde{p}$ and each predeclared penalty ρ_ℓ , the first-stage program solves

$$\min_{x,y,z} C_w(x) + \rho_\ell \sum_{d,t} z_{dt} \quad \text{s.t.} \quad z_{dt} \geq \pm(p_{dt}(x) - \tilde{p}_{dt}), \quad (16)$$

subject to (1)–(4). The absolute-value epigraph is standard [15]; its use for workload-ledger verification is specific here. C_w is workload cost, C_i is the generator-segment cost in (14), and $V(p)$ is dispatch value, so (16) changes only the objective. Six globally solved schedules $p^{(\ell)}$ share identical arrivals and constraints. A seventh matched-information feasible-quantile projection is retained as an external transfer comparator and is included only in the validation-only payment hull. The risk simplex below uses the six causal projections; the seven-profile hull is used only for the separate payment contract. For a hull profile, $\bar{p} = \sum_{\ell=1}^L \alpha_\ell p^{(\ell)}$, $\alpha \geq 0$, $\mathbf{1}^\top \alpha = 1$, remains feasible because the common scheduling set is convex.

The convex fit minimizes validation error plus total and daily-tail false-credit exposure, measured by conditional value at risk (CVaR) [20]. Let ℓ^* minimize contiguous-fold nRMSE and let B_h be the absolute daily false-credit exposure of that single feasible projection on validation day h . For a candidate simplex profile define $F_h(\alpha) = \sum_{n \in h} f_n$, where f_n is the positive-part false credit at sample n . The risk fit uses absolute MW-slot

exposure throughout; there is no ratio with a potentially zero true-credit denominator. With $m = \lceil (1 - \gamma)H \rceil$ and reserve $\beta \in (0, 1]$, solve

$$\min_{\alpha, f, \nu, u} \frac{\|\bar{p} - p^{0, \text{decl}}\|_2^2}{N s_p^2} + \lambda \|\alpha\|_2^2 + \omega_T \frac{\sum_{h=1}^H F_h}{S_T} \quad (17)$$

$$+ \omega_C \frac{\nu + m^{-1} \sum_{h=1}^H u_h}{S_C},$$

Equation (17) fixes both preferences inside the fit. Here $s_p^2 = \max\{N^{-1} \|p^{0, \text{decl}}\|_2^2, \epsilon\}$, $S_T = \max\{\sum_h B_h, \epsilon\}$, and $S_C = \max\{m^{-1} \sum_{h \in \text{tail}_\gamma} B_h, \epsilon\}$. Single-axis ablations remove one term and its constraint; the locked joint fit uses $\omega_T = \omega_C = 0.50$, $\gamma = 0.75$, and $\lambda = 10^{-8}$. The budgets are

$$\sum_{h=1}^H F_h(\alpha) \leq \beta \sum_{h=1}^H B_h, \quad (18)$$

$$\text{CVaR}_{0.75}(F_h(\alpha)) \leq \beta \text{CVaR}_{0.75}(B_h). \quad (19)$$

The positive part is represented without differentiating through a sort. For each event sample n and validation day h , introduce $f_n \geq 0$, a CVaR threshold $\nu \geq 0$, and tail slacks $u_h \geq 0$:

$$f_n \geq p_n^\alpha - p_n^{(\text{sim})} - r_n^{(\text{sim})}, \quad (20)$$

$$F_h(\alpha) = \sum_{n \in h} f_n, \quad (21)$$

$$F_h(\alpha) - \nu - u_h \leq 0, \quad (22)$$

$$\nu + \frac{1}{m} \sum_h u_h \leq \beta \text{CVaR}_{0.75}(B_h). \quad (23)$$

The linear CVaR epigraph in (23) uses the same m -day denominator as the empirical statistic and avoids sorting. Here $r_n^{(\text{sim})} = [p_n^0 - p_n^{(\text{sim})}]_+$; the implementation uses the equivalent threshold $p_n^{1, \text{sim}} + r_n^{(\text{sim})}$ and stores the identity audit. The constraints remain linear with the simplex and box bounds, while the objective is a convex quadratic program solved with CVXPY/Clarabel. Independent Karush–Kuhn–Tucker stationarity, epigraph, simplex, and primal residuals are recomputed after the solve. Four contiguous folds select β only when each training contract is feasible. A separately reported payment-contract projection targets $\bar{p}$ while imposing

$$p_{dt}^{\text{safe}} \leq p_{dt}^{(\ell^*)}. \quad (24)$$

The pointwise cap in (24) is the upper side of the contractual credit band; it is evaluated on every event slot before settlement and is not substituted for the scientific risk profile.

a) *Two-sided credit band.*: To exclude a zero-credit envelope, tie a lower band to the fitted target. Define $p_{dt}^{\text{floor}} = \min\{p_{dt}^{(\ell^*)}, \bar{p}_{dt}\}$, where $\bar{p}$ is the pre-projection validation fit. The event-slot constraints are

$$p_{dt}^{\text{floor}} - \varepsilon \leq p_{dt}^{\text{safe}} \leq p_{dt}^{(\ell^*)}, \quad (d, t) \in \mathcal{D} \times \mathcal{E}, \quad (25)$$

The release sets $\varepsilon = 0$; the lower side is therefore the exact pointwise minimum of the two independently formed profiles, and the selected single projection proves nonemptiness. Equations (18)–(19) control aggregate and daily-tail exposure, while the lower side in (25) regularizes under-credit and gives no unseen-trace guarantee. Realized overpayment is handled by (11).

b) *Decision-time information boundary.*: A gate removes all post-gate arrivals before refitting the matched-information

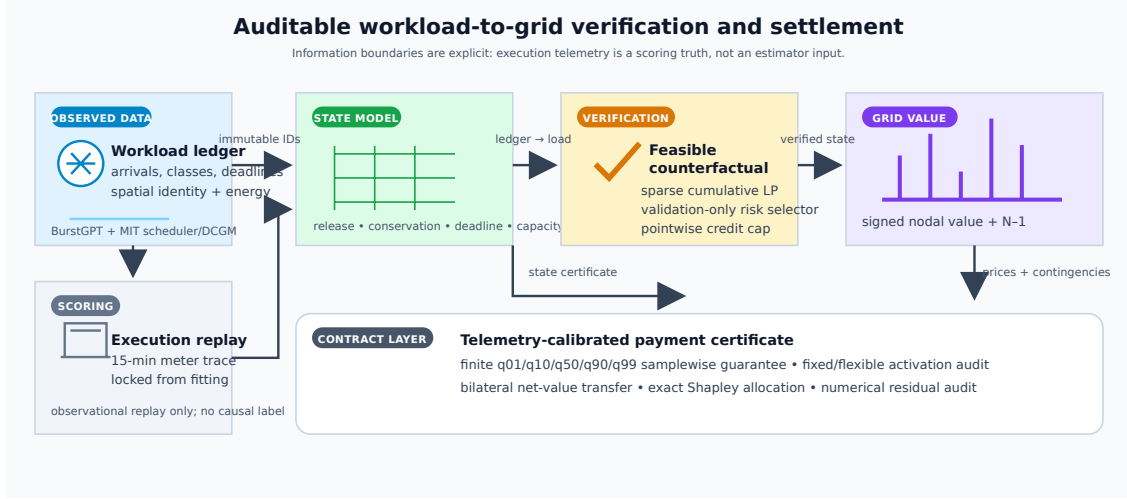


Fig. 1. Verification-to-settlement framework. Validation labels select the counterfactual risk contract; locked execution truth is used only for scoring. The right branch computes aggregate N-1 value and exact participant shares.

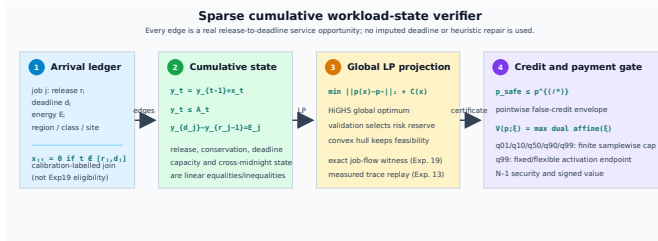


Fig. 2. Sparse cumulative verifier detail. Each admissible edge is a declared release-to-deadline service opportunity; the global projection, risk fit, and payment gate operate on the same feasible state and do not repair missing deadlines or execution records.

target. The locked protocol uses slot 62, 30 minutes before the first event slot, and a no-tariff LP freezes the contract baseline with a predeclared 3-MWh minimum of participating event-window service. A second LP applies the selected positive DR price to the same masked ledger while carrying unexpired state. Reserved future jobs cannot be credited, and the tariff plan is scored against the closed meter through (11). The independent execution trace is read only in the scoring pass.

To separate capacity planning from payment eligibility, the gate publishes a causal reserve envelope. For candidate quantile η and event day d ,

$$\hat{a}_{sk\tau}^{\text{res}}(d; \eta) = Q_{\eta}(\{a_{sk\tau}(h) : h < d\}), \quad \tau \geq t_{\text{gate}}, \quad (26)$$

Equation (26) uses earlier days only. The predeclared candidate set is $\{0.55, 0.60, 0.65, 0.70\}$ with a 1.0-MWh/day validation false-credit allowance; it selects $\eta = 0.55$ before locked scoring. The reserve is a planning envelope, and no reserved job enters payment before commitment.

The cumulative state has $O(|\mathcal{S}||\mathcal{K}||\mathcal{D}||\mathcal{T}|)$ nonzeros, versus quadratic explicit prefixes. Equation (5) retains every observed future arrival for the 512-slot deadline. Two lexicographic LPs (minimum L1, then operating cost) embed the day trajectory and are solved globally with HiGHS [21]. The resulting state-to-credit path is shown in Figs. 1 and 2; execution and interval audits are separate (Experiments 13–15).

C. Signed nodal and N-1 settlement

Let $\lambda^0 \in \partial V(p^0)$ be the nodal demand-value subgradient. Signed linear settlement is $(\lambda^0)^\top (p^0 - p^1)$, while (15) is the realized signed system-value benchmark. Convexity [15] gives

$$0 \leq (\lambda^0)^\top (p^0 - p^1) - \Delta V = V(p^1) - V(p^0) - (\lambda^0)^\top (p^1 - p^0). \quad (27)$$

Thus the linear rule upper-bounds exact value in the polyhedral market model; equality requires a common supporting subgradient.

For market-compatible remuneration, let $\mathcal{W}$ contain the complete response cycle from anticipatory shift through recovery. Its space-time remuneration is

$$\Pi^{\text{ST}} = \Delta t \sum_{t \in \mathcal{W}} \sum_d \lambda_{dt}^0 (p_{dt}^0 - p_{dt}^1). \quad (28)$$

This is the nodal virtual-link structure of [7], [8]; summing site payments reproduces (28).

Space-time energy remuneration is one component of the contracted DR product. Let $\mathcal{D}^{\text{part}}$ be the predeclared participating sites, $\mathcal{E}$ the event intervals, and r^{DR} the cleared service price. The signed delivered capacity service and total operator value are

$$Q^{\text{cap}} = \Delta t \sum_{d \in \mathcal{D}^{\text{part}}} \sum_{t \in \mathcal{E}} (p_{dt}^0 - p_{dt}^1), \quad (29)$$

$$G = r^{\text{DR}} Q^{\text{cap}} + \Pi^{\text{ST}}, \quad (30)$$

where p^0 is the minimum-cost no-event schedule on the same window. Equation (29) defines the bilateral capacity product and need not equal trace-anchored p^0 ; Π^{ST} debits spatial transfer and recovery. This signed service model follows demand-response mechanism design [22], [23].

Let x^1 and x^0 induce p^1 and p^0 . Define the participant opportunity cost $C_{\text{opp}} = C_w(x^1) - C_w(x^0)$ and surplus $S = G - C_{\text{opp}}$. With zero disagreement utilities, the symmetric Nash contract [24] is

$$z = \mathbb{I}\{S \geq 0\}, \quad P^B = z \left(C_{\text{opp}} + \frac{S}{2} \right), \quad u_{\text{dc}} = u_{\text{op}} = z \frac{S}{2}. \quad (31)$$

Equation (30) combines the capacity payment and signed space-time value. Equation (31) uses nonactivation as the outside option, keeps both utilities nonnegative, and is budget balanced. It is separate from ISO nodal remuneration and the exact dispatch-value benchmark. The complete-cycle ledger $\mathcal{W}$ includes pre-event increases and rebound, so workload conservation gives $\sum_{d,t \in \mathcal{W}} (p_{dt}^0 - p_{dt}^1) \Delta t = 0$ within solver tolerance.

Before payment comparison, the submitted, frozen-contract, and closed-meter reductions are intersected as follows:

$$q_{dt}^{\text{met}} = \min\{q_{dt}^{\text{sub}}, q_{dt}^{\text{cap}}\}, \quad G = r^{\text{DR}} \sum_{d,t \in \mathcal{E}} q_{dt}^{\text{met}} + \Pi^{\text{ST}}. \quad (32)$$

Equation (32) is evaluated after the submitted profile, contract, and meter are fixed. Experiment 12 stores the chain and budget residual (below 10^{-8} USD).

Security-aware settlement augments (14) with every finite non-islanding single-line outage. If $L_{\ell k}$ is the LODF from outage k to monitored line ℓ , then

$$-\bar{f}_{\ell} \leq (H_{\ell} + L_{\ell k} H_k)(C_g g - p) \leq \bar{f}_{\ell}, \quad \ell \neq k. \quad (33)$$

Equation (33) is the standard preventive N-1 LODF formulation [13]; every credible row uses native ratings. Islanding rows are excluded because a connected single-reference PTDF cannot represent separate islands, consistent with the DC/N-1 planning scope [25].

D. Scenario-robust N-1 payment certificate

The pointwise envelope controls energy but not network payment because spatial loads can activate different security constraints. Let $\mathcal{P} = \text{conv}\{p^{(1)}, \dots, p^{(L)}\}$ be the feasible-profile hull and $p^{\text{cap}} \in \mathcal{P}$ the validation-selected projection. Its penalty is separate from the payment-target penalty; the feasible-quantile projection is an external comparator. Let Ξ_5 be the 1st, 10th, 50th, 90th, and 99th calibration-validation GPU-energy ratios from independent telemetry [2]. For facility profile p_{dt}^{fac} , fixed demand P_d^{fix} , scale s_{dc} , and factor ξ ,

$$p_{dt}^{\text{dc},\xi} = s_{\text{dc}}(P_d^{\text{fix}} + \xi[p_{dt}^{\text{fac}} - P_d^{\text{fix}}]), \quad p_{dt}^{\text{fac}} \geq P_d^{\text{fix}}. \quad (34)$$

The scale uses the calibration-validation q_{99} flexible peak, keeping the fixed component invariant to ξ ; raw endpoints remain activation scenarios. The injection is $Mp_t^{\text{dc},\xi}$, with superscripts α and “cap” applying (34) to p^{α} and p^{cap} . For each settlement day, the payment-certified verifier defines $B_{\xi}^{\text{cap}} =$

$\Delta t \sum_{t \in \mathcal{E}} V_{N-1}(p_t^{\text{native}} + Mp_t^{\text{dc},\xi,\text{cap}})$ and solves

$$\begin{aligned} & \text{lexmin}_{\alpha, e, \eta, \{g_t^{\xi}\}} \left(\sum_{d,t \in \mathcal{E}} e_{dt}, -\eta \right) \\ & \text{s.t. } p^{\alpha} = \sum_{\ell=1}^L \alpha_{\ell} p^{(\ell)}, \\ & \alpha \geq 0, \quad \mathbf{1}^{\top} \alpha = 1, \quad 0 \leq \eta \leq 1, \\ & e_{dt} \geq \pm(p_{dt}^{\alpha} - p_{dt}^{\text{safe}}), \\ & g_t^{\xi} \text{ satisfies (14) and (33),} \\ & \text{with } p_t = p_t^{\text{native}} + Mp_t^{\text{dc},\xi,\alpha}, \\ & \quad t \in \mathcal{E}, \quad \xi \in \Xi_5, \\ & \Delta t \sum_{t \in \mathcal{E}} \sum_i C_i(g_{it}^{\xi}) \\ & \quad \leq (1 - \eta) B_{\xi}^{\text{cap}}, \quad \xi \in \Xi_5. \end{aligned} \quad (35)$$

The piecewise-linear epigraph is solved by two global LPs: trajectory error is minimized first, then the common fractional margin η . Feasibility is the N-1 upper bound and p^{cap} proves nonemptiness. Hence, for any $\xi \in \Xi_5$ and realized load $p^{1,\xi}$, daily payment obeys

$$\begin{aligned} & \sum_{t \in \mathcal{E}} \Delta t [V_{N-1}(p_t^{\text{native}} + Mp_t^{\text{dc},\xi,\alpha}) - V_{N-1}(p_t^{1,\xi})] \\ & \leq \sum_{t \in \mathcal{E}} \Delta t [V_{N-1}(p_t^{\text{native}} + Mp_t^{\text{dc},\xi,\text{cap}}) - V_{N-1}(p_t^{1,\xi})]. \end{aligned} \quad (36)$$

The realized cost cancels, so this is a samplewise certificate without execution truth. Every finite non-islanding contingency is imposed using standard SCED/LP primitives [12], [15]. The hull contains six first-stage projections and the matched feasible-quantile comparator. risk-constrained verifier is the validation-fitted simplex point under the two risk budgets; the separately stored payment contract targets it under the pointwise cap and is excluded from risk fitting.

The finite cap also has a set-valued certificate. Let $p^{\text{cap}} = p^{(\ell^*)}$, p^{pay} be the separately selected payment target, and $p^{\text{cert}} = p^{\alpha}$. For $\Delta V_{\xi}(p) = V_{N-1}(p, \xi) - V_{N-1}(p^{1,\xi})$ and $p_{\theta} = (1 - \theta)p^{\text{cap}} + \theta p^{\text{cert}}$, the interval is

$$\mathcal{I}_{\xi} = \left[\min_{0 \leq \theta \leq 1} \Delta V_{\xi}(p_{\theta}), \max\{\Delta V_{\xi}(p^{\text{cap}}), \Delta V_{\xi}(p^{\text{cert}})\} \right]. \quad (37)$$

The lower endpoint is one joint N-1 SCED LP with a shared segment parameter; convexity gives the endpoint maximum and the cap belongs to $\mathcal{I}_{\xi}$. Validation-frozen profiles and the realized counterfactual determine both endpoints; the interval width is uncertainty cost, while (35) remains the finite-scenario certificate.

E. Participant allocation

For participants N , let $v(S)$ be signed dispatch value when coalition S moves from baseline to actual demand. The exact allocation is

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)]. \quad (38)$$

This Shapley specialization in (38) uses signed grid value [26]; coalition-size grouping gives $\sum_i \phi_i = v(N)$ when

$v(\emptyset) = 0$. All 16 four-site and 256 eight-portfolio coalitions are enumerated exactly. For larger contracts, exchangeable site slices retain binomial multiplicities and reduce evaluation to $\prod_g(m_g + 1)$ count states (tested through 20 participants).

F. Analytical guarantees

These certificates delimit the logical scope. Under the interior and differentiability assumptions of (12), the threshold in (13) is exact; Experiment 1 checks all 56 cells. The workload set, simplex schedules, and two-sided envelope are feasible. For any execution c ,

$$\begin{aligned} [p_{dt}^{\text{safe}} - c_{dt}]_+ &\leq [p_{dt}^{(\ell^*)} - c_{dt}]_+, \\ [c_{dt} - p_{dt}^{\text{safe}}]_+ &\leq [c_{dt} - p_{dt}^{\text{floor}}]_+ + \varepsilon. \end{aligned}$$

Summation gives the false-credit and under-credit bounds without locked outcomes.

Base-case and stacked N-1 values are convex piecewise-affine functions of nodal demand; (27) gives the linear-value bound. Each dispatch embedded in (35) is feasible, so (36) holds for all $\xi \in \Xi_5$. Convexity in the continuous conversion factor puts the worst case on $[\xi_L, \xi_U]$ at an endpoint; (37) has the exact joint-LP lower endpoint. Experiments 9 and 15 use independent 40- and four-segment models.

The digest detects post-commit scheduler-tuple changes, while the execution join audits ordering and energy conservation. An optimum of (6) gives declaration-only runtime, capacity, and nameplate feasibility without aggregate rounding. With $B_{dj} = \mathbf{1}\{s_j = d\}$ and bus map M , the typed residuals are

$$\begin{aligned} r_j^{\text{job}} &= \sum_{\tau=r_j}^{d_j-1} u_{j\tau} - E_j^{\text{run}}, \\ r_{d\tau}^{\text{agg}} &= X_{d\tau} - \sum_j B_{dj} u_{j\tau}, \\ r_{b\tau}^{\text{map}} &= p_{b\tau} - P_{b\tau}^{\text{fix}} - \Delta t^{-1} \sum_d M_{bd} X_{d\tau}. \end{aligned}$$

The first residual uses the same E_j^{run} as (6); $\hat{E}_j$ remains diagnostic. Residuals at most ε identify the witness, checked in Experiments 22 and 27 before N-1 valuation; allocation checks $\sum_i \phi_i = v(N)$ separately.

IV. EXPERIMENTAL DESIGN

A. Data, preprocessing, and information boundary

BurstGPT supplies 5,188,507 successful requests and token service [1]; MIT Supercloud supplies scheduler submissions and independent Data Center GPU Manager (DCGM) energy. The full population contains 75,326 valid submissions and 71,128 positive-energy execution matches, of which 68,662 have valid intervals in the common 121-day tensor used by Experiments 1–13 [2]. Submission time defines arrivals and measured intervals define scoring truth. Chronological workload scaling uses the first 40 complete days. The GPU-power model uses 44,391 training jobs, 9,637 calibration-validation jobs, and 41,154 locked jobs; the calibration-validation ratios define the five scenarios Ξ_5 in (35) without synthetic replacement. Experiment 14 retains all 71,128 matches, whereas Experiment 19 uses every valid declaration with a training-only

aggregate entitlement. The submit-time energy-label fit contains 71,141 positive scheduler/DCGM pairs; 71,128 have valid measured intervals for replay, and the 13 label-only pairs cannot enter any decision. Experiment 27 forms the runtime-complete declaration witness, while execution joins open only for post-event scoring. The runtime window plus a precommitted 96-slot queue allowance defines the contract window, not a submit-to-completion interpretation of Slurm runtime, and membership is never outcome-filtered.

Because the traces lack utility bus locations, four-region coupling is a trace-driven benchmark: a predeclared feature-stratified round robin assigns class, GPU count, runtime, and submission time. Experiment 11 exhausts all 24 permutations, three penetrations, and two concentration controls; Experiment 18 uses fixed generator-bus strata at native-load multiplier 0.90, so its transfer result is conditional on that scenario. Calibration-validation ratios span 0.5827–5.3209 and the locked partition is a generalization audit. Equation (34) carries the fixed 6-MW/site component separately and calibrates payment-network scale on q_{99} at a predeclared 4% peak penetration; both raw endpoints remain activation-eligible under that N-1-feasible scale. The separate flexible-only capacity-safe factor is retained as a planning diagnostic and is not an endpoint clip. The Power Grid Library (PGLib) and MATPOWER/PYPOWER provide topology, ratings, and costs [12], [27]; RTS-24 follows the published case [28].

B. Baselines, parameters, and inference

Experiment 2 compares meter/statistical baselines with complete-ledger quantile/feasible projections, synthetic control, tail-risk counterfactual, the selected single projection, and risk-constrained verifier; payment-certified verifier is evaluated separately in Experiments 9 and 15. Feature equality is audited while execution truth is withheld. Deferral windows are 0, 4, and 512 slots; each region has a 118-MW flexible nameplate and 6-MW fixed component. Six projection penalties and four risk reserves are selected by nested contiguous folds; normalized total/CVaR preferences and the five-point stress grid are validation-only. The gate uses slot 62 and commits 3 MWh; payment uses an independent 40-segment evaluator. Moving three-day bootstrap replicates and all 2^{18} block-sign assignments use Holm correction [29], [30]. Sparse-system size grows from 5,760 variables/13,424 nonzeros at 96 slots to 36,480/85,488 at 608 slots (0.039–0.212 s). All release runs clamp BLAS and HiGHS to one numerical-backend thread per worker; the bounded pools use at most 12 workers and remain below the project-wide 20-core ceiling. Every long panel writes a schema- and profile-validated checkpoint before a resumed cell is accepted.

The model-out check uses polar alternating-current optimal power flow (AC OPF) with active/reactive balance, voltage/reactive-generation bounds, and apparent-power ratings [12]. After the intact solve, each credible outage fixes every non-reference active output at its intact value, $P_{Gi}^{(k)} = P_{Gi}^{(0)}$, while the reference active balance, reactive outputs, and voltages remain recourse. Apparent-power, voltage, and fixed-plan deviations are checked after every solve; non-reference bounds

are the intact output $\pm 10^{-8}$ MW with a 10^{-6} -MW margin, and the largest deviation is 2.52×10^{-8} MW.

Experiments 1–2 test incentives/workload verification; 3–9 test signed settlement, transfer, risk, and N–1 value; 10–12 test AC/spatial transfer and continuous-cycle remuneration; 13–16 provide measured replay, indexed witness, conversion intervals, and reconciliation. Experiments 17–25 freeze the gate, verify shared-plan AC feasibility, audit scale/coupling, replay an independent event, and enumerate all RTS-24 outages. Experiment 27 adds the runtime-complete common witness, and Experiment 28 audits the held-out risk tail with a fixed block bootstrap. Comparisons use the same 54 locked days unless a deterministic network or ledger rule is stated. Experiment 26 independently recomputes the declaration-to-network residuals and checks the common witness digest, source hashes, and role labels of the risk, payment, and outage artifacts.

V. RESULTS

A. Incentive boundary and independent verification

Experiment 1 solves the entire 8×7 DR-price/call-probability grid. The dual threshold (13) agrees with the independently solved strategic program in all 56 cells. At \$150/MWh and $q = 0.35$, strategic reference scheduling inflates the event baseline by 8.223 MWh. Of 55.452 MWh credited, 44.458 MWh is unsupported by physical reduction, giving a false-credit ratio of 0.802. The profitable region follows the derived marginal boundary and does not use a fitted classifier; the complete grid and its phase diagram are retained in the experiment output.

Experiment 2 evaluates twelve methods on the locked 54-day panel (Table I). The joint total-plus-daily-CVaR fit has $nRMSE/F_1$ /false credit/under-credit of 1.123/0.373/0.303/37.994 MWh/day. The selected single feasible projection has 0.943/0.463/2.470/26.812. Thus risk control deliberately exchanges point-estimate and under-credit performance for a lower unsupported-credit exposure; it is not a dominance claim. The validation certificate, fixed before the locked split, reduces total false exposure from 58.104 to 4.507 MW-slots under a 14.526 MW-slot budget and reduces its absolute daily $CVaR_{0.75}$ from 12.993 to 0.966 under a 3.248 budget. The total-budget-only and CVaR-only ablations give 1.122/0.378/0.283 and 1.121/0.384/0.264 for $nRMSE/F_1$ /false credit, respectively. The joint fit is therefore identified by the simultaneous total and tail constraints rather than by a post-solve clipping rule. Its convex QP has Clarabel KKT stationarity residual 3.52×10^{-10} , primal residual 8.0×10^{-15} , and zero linear-epigraph residual. The independent meter replay is reported separately: RiskSafe has $nRMSE$ 0.455 and MAE 4.813 MW, compared with 0.716 and 7.038 MW for the single projection. No trace-meter value is used in fitting or reserve selection.

The independent trace-meter replay is a deployment check: event-window MAE is 1.909/5.302/4.813 MW for the feasible-quantile comparator/tail-risk counterfactual/risk-constrained verifier over the same 54 days (Experiment 20), separate from mechanism-isolation scoring. Closest domain translations use identical arrivals, deadlines, capacities, event slots, and days: the rolling-horizon ledger gives $nRMSE$ /false-credit/ F_1 0.943/2.470/0.463, the batch-flexibility translation

TABLE I
LOCKED 54-DAY MECHANISM-ISOLATION RESPONSE RESULTS (DAILY MEANS).

Method	nRMSE	False MWh	Under-credit	F_1
High-5-of-10	1.586	41.220	11.169	0.572
Ridge	1.038	21.906	25.226	0.545
Gradient Boosting	1.011	20.977	26.401	0.534
Extra Trees	1.090	26.629	18.176	0.597
Metadata Gradient Boosting	0.774	9.183	21.528	0.640
Ex-post Metadata Gradient Boosting	0.750	9.171	20.817	0.643
Ex-post Quantile Gradient Boosting	1.337	0.591	54.391	0.237
Synthetic Control	0.782	9.394	21.678	0.676
Feasible Quantile Projection	1.344	0.098	52.647	0.225
Single Feasible Projection	0.943	2.470	26.812	0.463
Tail-Risk Feasible Counterfactual	1.121	0.264	36.562	0.384
Risk-Constrained Convex Verifier	1.123	0.303	37.994	0.373

TABLE II
BASE-CASE AND COMPLETE N–1 SETTLEMENT ERRORS (USD/DAY).

Panel	Settlement	Trace MAE	Workload MAE	Trace median/overpay	Workload median/overpay
Base case	Nodal gross	360.580	421.677	–	–
Base case	Nodal exact net value	151.719	455.865	–	–
N–1	Base exact	139.221	167.367	79.79/130.42	121.01/107.69
N–1	N–1 linear	2.208	150.461	0.60/1.91	113.43/20.38
N–1	N–1 exact	0.990	150.955	0.87/0.01	113.43/20.18

gives 1.357/0/0.183, and the coupled-service translation gives 1.376/0/0.111. These controls translate the published structures into the same exact ledger and are labelled translations rather than software reproductions [5], [10]. The ablation panel contains 810 globally solved schedules; its binding-share and marginal-cost diagnostics are reported in the archived certificate.

B. Nodal value and spatial response

Experiment 3 crosses thirteen counterfactuals with five settlement mechanisms; Table II reports the trace-anchored locked rows. Exact signed nodal value has MAE \$151.719 versus \$360.580 for nodal gross response, with workload errors \$455.865 and \$421.677. Signed netting reduces trace error by \$208.190/day (Holm-adjusted $p = 3.89 \times 10^{-4}$); uniform-to-nodal pricing changes it by $-\$1.463$ ($p = 0.0118$), and linear-to-exact value by $\$0.671$ ($p = 0.8026$). All 5,616 interval/counterfactual instances satisfy (27), with maximum residual 2.03×10^{-11} dollars.

Experiment 4 applies the predeclared largest gross-minus-net offset (day 87), exposing the rebound component that positive-only local accounting omits.

C. Cross-network and N–1 security evidence

Experiment 5 evaluates 6,480 four-network outcomes; exact value improves the trace-anchored error in all 12 cells (mean \$1.913/day; eight Holm-adjusted tests below 0.05) with a \$0.050/day workload change. Experiment 6 solves 810 capacity/deadline schedules (binding shares 0.031–0.911; false credit 2.374–2.740 MWh). Experiment 7 enumerates 16 four-site and 256 eight-site coalitions (budget residual 5.68×10^{-14} dollars) plus 1,296 exact 20-participant states. Experiment 8 enforces (33) for all 37 RTS-24 non-islanding outages; trace/workload MAE is \$139.221/\$2.208/\$0.990 and \$167.367/\$150.461/\$150.955 for base exact, N–1 linear, and N–1 exact, with 1.000 p.u. maximum loading.

Table III summarizes the complete replay scope.

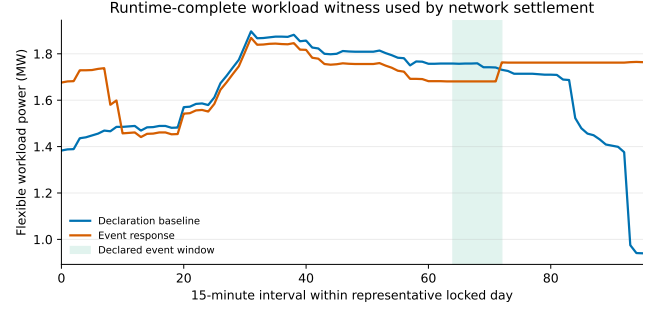
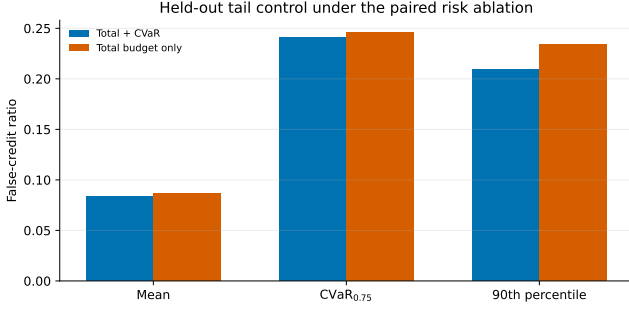


Fig. 3. Result evidence for the two coupled certificates. Left: held-out absolute daily false-credit exposures for the joint total-plus-CVaR fit and the total-budget-only ablation, including the empirical upper tail. Right: declaration baseline and event response generated by the same runtime-complete job witness on a locked day; the shaded interval is the predeclared event window. The settlement ledger retains the full 96-slot day and flags the eight event slots.

Experiment 9 tests (36) on 54 programs with a validation-frozen seven-profile hull; contiguous-fold nRMSE selects the cap reference (Projection $\rho = 10$) and validation N-1 MAE selects the payment target (Projection $\rho = 3$). The declared conversion factors are $q_{01} = 0.583$, $q_{10} = 0.687$, $q_{50} = 0.970$, $q_{90} = 1.349$, and $q_{99} = 5.321$; every certificate violation is below 3×10^{-11} USD. The q50/q99 MAE (USD/day) for PaymentSafe/single/feasible-quantile/RiskSafe is 56.858/321.660, 40.675/231.475, 55.360/312.933, and 55.877/317.475, respectively; corresponding overpayment is at most 1.087/5.118 USD/day for the feasible-quantile comparator and below 0.031/0.024 USD/day for the single projection. Their moving-block intervals and the unseen 0.80/1.20 transfer panel are stored with the locked artifact.

Experiment 10 replaces the linear network with AC OPF [12]: all 864 connected-base and 5,400 corrective IEEE-9/14 cells are feasible, with zero deviation in 3,888 shared-plan cells (0.880/0.907/0.935 p.u. at 3/6/9%). Experiment 18 extends the fixed-active plan to four networks and 9,648 admissible contingency solves plus 144 intact solves; all remaining cells have zero voltage violation and maximum loading 1.000 p.u. Experiment 11 averages 26 assignments across three penetrations, four counterfactuals, and 54 days; its 3/6/9% MAE is \$101.155/\$203.361/\$303.195 for RiskSafe, \$105.250/\$211.555/\$315.490 for single, and \$81.613/\$164.249/\$244.481 for feasible quantile. Experiment 16 audits 44,391/9,637/41,154 training, calibration-validation, and locked GPU records; MAE/RMSE/ R^2 are 10.803/14.463/0.8956 and 13.543/20.075/0.8196 W/W/1. Experiment 12 reports event-window MAE \$26.937/\$34.166/\$8.659/\$5.325 and complete-cycle residual \$0.621/\$0.569/\$0.104/\$0 for feasible-quantile, payment-certified, RiskSafe, and single.

D. Independent trace and deployment-boundary audits

The immutable scheduler/DCGM replay gives a 71,128-job residual below 10^{-17} MWh; no utility-event label exists. Experiment 17 quantifies the information cost: the complete-ledger risk profile has nRMSE/false response/ F_1 1.123/0.303/0.373, while the slot-62 gate with post-gate arrivals masked gives 1.366/0.494/0.080. The gate is fixed before scoring, removes post-gate arrivals from every decision LP, and requires 3 MWh

TABLE III
DECLARED INFORMATION-BOUNDARY AND SECURITY-REPLAY COVERAGE.
THE INDEPENDENT DECLARATION CHECKS ARE COMPUTED BEFORE
LOCKED SCORING.

Panel	Cells and certificate
Independent declaration replay	777 information and independence checks; nRMSE 0.0210, $F_1 = 0.529$
Frozen-profile DC N-1 (RTS-24)	864 cells; 37 outages/cell; 1.0000 p.u.
Shared-plan AC (IEEE-9)	3,888; 0.880/0.907/0.935 p.u.
Corrective AC (IEEE-9/14)	864 base + 5,400 outage solves; all feasible
Admissible fixed-plan AC N-1 (four networks)	9,648 contingency + 144 intact solves; all certified

of committed event service on all 54 locked days. Experiment 20 gives trace-meter MAE 4.813 MW for the risk profile (1.909 for the feasible-quantile comparator). Experiment 23 enumerates every declaration-feasible start under a distinct tariff; its common witness gives nRMSE 0.0210, false response 0.0194 MWh/day, and $F_1 = 0.529$ over 54 days, while the independently recomputed policy is an exact $F_1 = 1.000$ reference. The certificate verifies no future arrivals or execution telemetry enter the policy and makes no field-causal claim. Table III records the boundary.

E. Indexed workload counterfactual

Experiment 27 assigns each of 75,326 valid submissions a runtime-complete contiguous block at the declared GPU nameplate. The indexed Exp19 witness has 13,198,247 job-slot variables; Exp27 enumerates every admissible start in each declaration window. The nameplate and calibrated central entitlements are 1,972.115 and 0.570 MWh, with both residuals below 10^{-11} MWh and 28.595 MWh minimum site slack (Table IV); the event-window reduction is 4.68225 MWh. Signed N-1 settlement covers 5,184 full-day cells (432 event cells), each with 37 finite contingencies. Experiment 25 validates the 71,128/75,326 execution join and its 79-job binding cohort; Experiment 24 independently covers 864 frozen-profile cells and 31,968 contingency solves. Experiment 26 recomputes the job-to-region and region-to-bus residuals, the central-energy/risk bridge, and the common digest before settlement; its typed lineage certificate passes. The risk-to-executable audit uses the locked risk-constrained verifier profile to form a fixed executable contract. Because the statistical profile and the declared GPU nameplate are calibrated on

TABLE IV
EXACT INDEXED JOB-LEVEL AND BINDING-CAPACITY CERTIFICATES.

Quantity	Value
Submitted / execution-matched jobs; admissible starts	75,326 / 71,128; 13,198,247
Nameplate upper / central declared runtime energy (MWh)	1,972.115 / 0.570
Nameplate baseline / response event energy; event reduction (MWh)	165.436 / 160.754; 4.68225
Maximum aggregate energy residual; minimum site slack (MWh)	7.8×10^{-12} ; 28.595
Binding stress cohort / capacity / service floor (jobs/MW/MWh)	79 / 0.00016037 / 0.00108
Job-to-network residual; full-day / event settlement cells	0; 5,184 / 432
Risk-priced executable bridge mean nRMSE (upper / central)	0.02601 / 8.41×10^{-6}
DC N-1 profile panel; RTS-24 outages / cells / evaluations	37 / 864 / 31,968
Maximum counterfactual post-contingency loading (p.u.)	1.0000

different absolute scales, the flexible part is multiplied once by a global factor determined from locked-day submit-time declarations; the raw profile remains in the certificate. Regional slot prices are then fixed from that scaled shape and every declaration-feasible start is enumerated. Its upper-capacity and central-energy realizations are stored with daily residuals and the same submission digest, so the aggregate fit is connected to an integer, contiguous job state before any network solve. Full-day settlement keeps negative network-value cells as signed debits; the event-slot subset is reported only as an event diagnostic.

VI. CONCLUSION

This paper couples a provenance-bound workload verifier with signed nodal settlement. On locked data, the joint profile reaches $\text{nRMSE}/\text{false credit}/F_1 = 1.123/0.303/0.373$ MWh/day, with both the aggregate and absolute daily-CVaR budgets checked independently (Fig. 3). The runtime-complete declaration witness covers all 75,326 submissions, conserves 1,972.115 MWh of declared runtime energy, shifts 4.68225 MWh from the event window, and carries one digest through 5,184 full-day RTS-24 settlement cells (432 event cells) and 37 finite N-1 outages per cell; the signed payable settlement is 277.806 USD, with negative network-value cells retained as signed debits before the pointwise meter cap. The risk-to-executable bridge has mean nRMSE 0.0260 for the nameplate envelope. The independent declaration replay confirms the response under a separate exact tariff policy. Fixed demand remains outside the flexible conversion, and deployment requires signed scheduler, telemetry, and topology attestation; field-causal validation remains a separate operational step. The aggregate risk fit is therefore reported as a statistical contract view, while the runtime-complete witness is the executable physical state used for network settlement.

REFERENCES

- [1] Y. Wang, Y. Chen, Z. Li, X. Kang, Y. Fang, Y. Zhou, Y. Zheng, Z. Tang, X. He, R. Guo, X. Wang, Q. Wang, A. C. Zhou, and X. Chu, "BurstGPT: A real-world workload dataset to optimize large language model serving systems," in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pp. 5831–5841, 2025.
- [2] S. Samsi, M. L. Weiss, D. Bestor, B. Li, M. Jones, A. Reuther, *et al.*, "The mit supercloud dataset," in *2021 IEEE High Performance Extreme Computing Conference*, pp. 1–8, 2021.
- [3] California Independent System Operator, "California iso baseline accuracy work group final proposal," tech. rep., California ISO, 2017.
- [4] X. Wang and W. Tang, "Modeling and analysis of baseline manipulation in demand response programs," *IEEE Transactions on Smart Grid*, vol. 13, no. 2, pp. 1178–1186, 2022.
- [5] Y. Cao, M. Cheng, S. Zhang, H. Mao, P. Wang, C. Li, Y. Feng, and Z. Ding, "Data-driven flexibility assessment for internet data center towards periodic batch workloads," *Applied Energy*, vol. 324, p. 119665, 2022.
- [6] Y. Cao, F. Cao, Y. Wang, J. Wang, L. Wu, and Z. Ding, "Managing data center cluster as non-wire alternative: A case in balancing market," *Applied Energy*, vol. 360, p. 122769, 2024.
- [7] W. Zhang, L. A. Roald, A. A. Chien, J. R. Birge, and V. M. Zavala, "Flexibility from networks of data centers: A market clearing formulation with virtual links," *Electric Power Systems Research*, vol. 189, p. 106723, 2020.
- [8] W. Zhang and V. M. Zavala, "Remunerating space-time, load-shifting flexibility from data centers in electricity markets," *Applied Energy*, vol. 326, p. 119930, 2022.
- [9] W. Zhang, L. A. Roald, and V. M. Zavala, "Exploring the impacts of power grid signals on data center operations using a receding-horizon scheduling model," *IEEE Transactions on Sustainable Computing*, vol. 8, no. 2, pp. 245–256, 2023.
- [10] M. Chen, C. Gao, M. Shahidehpour, Z. Li, S. Chen, and D. Li, "Internet data center load modeling for demand response considering the coupling of multiple regulation methods," *IEEE Transactions on Smart Grid*, vol. 12, no. 3, pp. 2060–2076, 2021.
- [11] B. Stott, J. Jardim, and O. Alsac, "DC power flow revisited," *IEEE Transactions on Power Systems*, vol. 24, no. 3, pp. 1290–1300, 2009.
- [12] R. D. Zimmerman, C. E. Murillo-Sanchez, and R. J. Thomas, "MATPOWER: Steady-state operations, planning, and analysis tools for power systems research and education," *IEEE Transactions on Power Systems*, vol. 26, no. 1, pp. 12–19, 2011.
- [13] D. A. Tejada-Arango, P. Sánchez-Martín, and A. Ramos, "Security constrained unit commitment using line outage distribution factors," *IEEE Transactions on Power Systems*, vol. 33, no. 1, pp. 329–337, 2018.
- [14] National Institute of Standards and Technology and Q. Dang, "Secure hash standard," Tech. Rep. FIPS PUB 180-4, National Institute of Standards and Technology, 2015.
- [15] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge University Press, 2004.
- [16] A. E. Hoerl and R. W. Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," *Technometrics*, vol. 12, no. 1, pp. 55–67, 1970.
- [17] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *The Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.
- [18] P. Geurts, D. Ernst, and L. Wehenkel, "Extremely randomized trees," *Machine Learning*, vol. 63, pp. 3–42, 2006.
- [19] A. Abadie, A. Diamond, and J. Hainmueller, "Synthetic control methods for comparative case studies: Estimating the effect of california's tobacco control program," *Journal of the American Statistical Association*, vol. 105, no. 490, pp. 493–505, 2010.
- [20] R. T. Rockafellar and S. Uryasev, "Optimization of conditional value-at-risk," *The Journal of Risk*, vol. 2, no. 3, pp. 21–41, 2000.
- [21] Q. Huangfu and J. A. J. Hall, "Parallelizing the dual revised simplex method," *Mathematical Programming Computation*, vol. 10, no. 1, pp. 119–142, 2018.
- [22] B. Satchidanandan, M. Roozbehani, and M. A. Dahleh, "A two-stage mechanism for demand response markets," *IEEE Control Systems Letters*, vol. 7, pp. 49–54, 2023.
- [23] M. Chen, C. Gao, M. Shahidehpour, and Z. Li, "Incentive-compatible demand response for spatially coupled internet data centers in electricity markets," *IEEE Transactions on Smart Grid*, vol. 12, no. 4, pp. 3056–3069, 2021.
- [24] J. John F. Nash, "The bargaining problem," *Econometrica*, vol. 18, no. 2, pp. 155–162, 1950.
- [25] North American Electric Reliability Corporation, "TPL-001-5.1: Transmission system planning performance requirements," tech. rep., North American Electric Reliability Corporation, 2020.
- [26] L. S. Shapley, "A value for n-person games," in *Contributions to the Theory of Games, Volume II* (H. W. Kuhn and A. W. Tucker, eds.), pp. 307–317, Princeton University Press, 1953.
- [27] S. Babaeinejadsarookolae, A. Birchfield, R. D. Christie, C. Coffrin, C. DeMarco, R. Diao, *et al.*, "The power grid library for benchmarking ac optimal power flow algorithms," *arXiv preprint arXiv:1908.02788*, 2019.
- [28] C. Grigg, P. Wong, P. Albrecht, R. Allan, M. Bhavaraju, R. Billinton, Q. Chen, C. Fong, S. Haddad, S. Kuruganty, *et al.*, "The IEEE reliability test system-1996," *IEEE Transactions on Power Systems*, vol. 14, no. 3, pp. 1010–1020, 1999.
- [29] H. R. Künsch, "The jackknife and the bootstrap for general stationary observations," *The Annals of Statistics*, vol. 17, no. 3, pp. 1217–1241, 1989.
- [30] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.